COMP3011 Climate & Urban Environment Statistics API

1.0.0 OAS 3.1

/openapi.json

Stores historical city observations (temperature + PM2.5) and provides analytics endpoints.

default

GET / Root

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header. **Example Value** Schema `"string"`	*No links*

GET /cities List Cities

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	Successful Response Media type **application/json** Controls `Accept` header. **Example Value** Schema	*No links*

```
[
  {
    "id": 0,
    "name": "string",
    "country": "string",
    "lat": 0,
    "lon": 0
  }
]
```

POST /cities Create City

Parameters

Try it out

No parameters

Request body required

application/json

Example Value Schema

```
{
  "name": "string",
  "country": "string",
  "lat": -90,
  "lon": -180
}
```

Responses

Code	Description	Links
201	Successful Response Media type **application/json** Controls `Accept` header. **Example Value** Schema	*No links*

Code	Description	Links


```
{
  "id": 0,
  "name": "string",
  "country": "string",
  "lat": 0,
  "lon": 0
}
```


422 Validation Error *No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


GET `/cities/compare` Compare Cities

Parameters Try it out

Name	Description
city1 * required `integer` *(query)*	city1 *minimum: 1*
city2 * required `integer` *(query)*	city2 *minimum: 1*
start * required `string($date)` *(query)*	start
end * required `string($date)` *(query)*	end

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value Schema


```
{
  "city_1": {
    "city_id": 0,
    "city_name": "string",
    "country": "string",
    "start": "2026-03-10",
    "end": "2026-03-10",
    "count_days": 0,
    "avg_temp_c": 0,
    "avg_pm25": 0,
    "max_temp_c": 0,
    "max_pm25": 0
  },
  "city_2": {
    "city_id": 0,
    "city_name": "string",
    "country": "string",
    "start": "2026-03-10",
    "end": "2026-03-10",
    "count_days": 0,
    "avg_temp_c": 0,
    "avg_pm25": 0,
    "max_temp_c": 0,
    "max_pm25": 0
  },
  "hotter_city": "string",
  "more_polluted_city": "string",
  "temp_diff_avg": 0,
```


422	Validation Error	*No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

GET /cities/{city_id} Get City

Parameters

Try it out

Name	Description
city_id * required integer (path)	city_id

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema	No links
422	Validation Error Media type application/json Example Value Schema	No links

```
{
  "id": 0,
  "name": "string",
  "country": "string",
  "lat": 0,
  "lon": 0
}
```

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

PUT /cities/{city_id} Update City

Parameters

Try it out

Name	Description
city_id * required integer (path)	city_id

Request body required

application/json

Example Value Schema

```
{
  "name": "string",
  "country": "string",
  "lat": -90,
  "lon": -180
}
```

Responses

Code	Description	Links

200	Successful Response
	Media type
	application/json
	Controls Accept header.
	Example Value Schema
	{
	"id": 0,
	"name": "string",
	"country": "string",
	"lat": 0,
	"lon": 0
	}
		No links

422	Validation Error Media type application/json Example Value Schema { "detail": [	*No links*

Code	Description	Links


```
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


DELETE `/cities/{city_id}` Delete City

Parameters Try it out

Name	Description
city_id * required `integer` *(path)*	city_id

Responses

Code	Description	Links
204	Successful Response	*No links*
422	Validation Error	*No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

GET /observations Query Observations

Parameters

Try it out

Name	Description
city_id * required integer *(query)*	city_id *minimum: 1*
start string | (string | null)($date) *(query)*	start
end string | (string | null)($date) *(query)*	end
limit integer *(query)*	*Default value* : 200 200 *maximum: 2000* *minimum: 1*
offset integer *(query)*	*Default value* : 0 0 *minimum: 0*
sort string *(query)*	*Default value* : obs_date obs_date *pattern: ^(obs_date|temp_c|pm25)$*
order string *(query)*	*Default value* : asc asc *pattern: ^(asc|desc)$*

Responses

Code	Description	Links
200	Successful Response Media type **application/json** Controls `Accept` header.	*No links*

Code	Description	Links

Example Value Schema


```
[
  {
    "id": 0,
    "city_id": 0,
    "obs_date": "2026-03-10",
    "temp_c": 0,
    "pm25": 0
  }
]
```


422 Validation Error *No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


GET `/cities/{city_id}/summary` City Summary

Parameters Try it out

Name	Description
city_id * required `integer` *(path)*	city_id
start * required `string($date)` *(query)*	start
end * required `string($date)` *(query)*	end

Responses

Code	Description	Links
200	Successful Response Media type **application/json** Controls `Accept` header. **Example Value** Schema	*No links*
422	Validation Error Media type **application/json** **Example Value** Schema	*No links*

```
{
  "city_id": 0,
  "start": "2026-03-10",
  "end": "2026-03-10",
  "count_days": 0,
  "avg_temp_c": 0,
  "min_temp_c": 0,
  "max_temp_c": 0,
  "avg_pm25": 0,
  "min_pm25": 0,
  "max_pm25": 0
}
```

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

GET `/cities/{city_id}/trend` City Trend

Parameters

Try it out

Name	Description
city_id * required `integer` *(path)*	city_id
metric	*Default value* : temp_c

Name	Description
string (query)	temp_c *pattern: ^(temp_c|pm25)$*
start * required string($date) *(query)*	start
end * required string($date) *(query)*	end
window integer *(query)*	*Default value* : 7 7 *maximum: 60* *minimum: 1*

Responses

Code	Description	Links
200	Successful Response Media type **application/json** Controls `Accept` header. **Example Value** Schema	*No links*
422	Validation Error Media type **application/json** **Example Value** Schema	*No links*

```
{
  "city_id": 0,
  "metric": "string",
  "start": "2026-03-10",
  "end": "2026-03-10",
  "window": 0,
  "points": [
    {
      "date": "2026-03-10",
      "value": 0,
      "moving_avg": 0
    }
  ]
}
```

Code	Description	Links


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


GET `/cities/{city_id}/anomalies` City Anomalies

Parameters Try it out

Name	Description
city_id * required integer *(path)*	city_id
metric string *(query)*	*Default value* : temp_c temp_c *pattern: ^(temp_c|pm25)$*
start * required string($date) *(query)*	start
end * required string($date) *(query)*	end
method string *(query)*	*Default value* : zscore zscore *pattern: ^(zscore|rolling_zscore|iqr)$*
threshold number *(query)*	*Default value* : 2 2 *maximum: 10* *minimum: 0.5*
window integer *(query)*	*Default value* : 14 14 *maximum: 120* *minimum: 3*

Name	Description
k number *(query)*	*Default value* : 1.5 1.5 *maximum: 10* *minimum: 0.5*

Responses

Code	Description	Links
200	Successful Response Media type **application/json** Controls `Accept` header. **Example Value** Schema	*No links*

```
{
  "city_id": 0,
  "metric": "string",
  "start": "2026-03-10",
  "end": "2026-03-10",
  "method": "zscore",
  "window": 0,
  "threshold": 0,
  "k": 0,
  "mean": 0,
  "std": 0,
  "q1": 0,
  "q3": 0,
  "iqr": 0,
  "anomalies": [
    {
      "date": "2026-03-10",
      "value": 0,
      "score": 0,
      "direction": "high"
    }
  ]
}
```

422	Validation Error Media type **application/json** **Example Value** Schema	*No links*

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
```

Code	Description	Links


```
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


GET `/cities/{city_id}/risk-score` City Risk Score

Parameters

Try it out

Name	Description
city_id * required `integer` *(path)*	city_id
start * required `string($date)` *(query)*	start
end * required `string($date)` *(query)*	end

Responses

Code	Description	Links
200	Successful Response Media type **application/json** Controls `Accept` header. **Example Value** Schema	*No links*


```
{
  "city_id": 0,
  "start": "2026-03-10",
  "end": "2026-03-10",
  "risk_score": 0,
  "label": "Low",
  "components": {
    "additionalProp1": 0,
    "additionalProp2": 0,
    "additionalProp3": 0
  }
}
```

Code	Description	Links
422	Validation Error	*No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


GET `/cities/{city_id}/regimes` City Regimes

Parameters Try it out

Name	Description
city_id * required `integer` *(path)*	city_id
metric `string` *(query)*	*Default value* : temp_c temp_c *pattern: ^(temp_c|pm25)$*
start * required `string($date)` *(query)*	start
end * required `string($date)` *(query)*	end
window `integer` *(query)*	*Default value* : 14 14 *maximum: 120* *minimum: 3*

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value Schema


```
{
  "city_id": 0,
  "metric": "string",
  "start": "2026-03-10",
  "end": "2026-03-10",
  "window": 0,
  "points": [
    {
      "date": "2026-03-10",
      "value": 0,
      "rolling_std": 0,
      "regime": "Stable"
    }
  ]
}
```


422	Validation Error	*No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


GET `/cities/{city_id}/insights` City Insights

Parameters

Try it out

Name	Description
city_id * required integer *(path)*	city_id
start * required string($date) *(query)*	start
end * required string($date) *(query)*	end

Responses

Code	Description	Links
200	Successful Response Media type **application/json** Controls `Accept` header. **Example Value** Schema	*No links*
422	Validation Error Media type **application/json** **Example Value** Schema	*No links*

```
{
  "city_id": 0,
  "city_name": "string",
  "start": "2026-03-10",
  "end": "2026-03-10",
  "insight": "string"
}
```

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
```

Code	Description	Links

```
  ]
}
```

GET `/cities/rankings` City Rankings

Parameters Try it out

Name	Description
metric * required `string` *(query)*	metric *pattern: ^(temp_c|pm25|risk)$*
start * required `string($date)` *(query)*	start
end * required `string($date)` *(query)*	end
limit `integer` *(query)*	*Default value* : 10 10 *maximum: 50* *minimum: 1*

Responses

Code	Description	Links
200	Successful Response Media type **application/json** Controls `Accept` header. **Example Value** Schema	*No links*

```
{
  "metric": "string",
  "start": "2026-03-10",
  "end": "2026-03-10",
  "limit": 0,
  "rankings": [
    {
      "city_id": 0,
      "city_name": "string",
      "country": "string",
```

Code	Description	Links


```
      "value": 0
    }
  ]
}
```


422 Validation Error *No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


Schemas

AnomalyOut > Expand all `object`

AnomalyPoint > Expand all `object`

CityCompareOut > Expand all `object`

CityCompareStats > Expand all `object`

CityCreate > Expand all `object`

CityInsightsOut > Expand all `object`

CityOut > Expand all `object`

CityRankingItem > Expand all `object`

CityRankingOut > Expand all `object`

CitySummaryOut > Expand all `object`

CityUpdate > Expand all `object`

HTTPValidationError ^ Collapse all `object`

`detail` > Expand all `array<object>`

ObservationOut > Expand all `object`

RegimePoint > Expand all `object`

RegimesOut > Expand all `object`

RiskScoreOut > Expand all `object`

TrendOut > Expand all `object`

TrendPoint > Expand all `object`

ValidationError > Expand all `object`